

Intrinsic invariants of $(3, 4)$ -cuspidal edges

Kaoru Iino

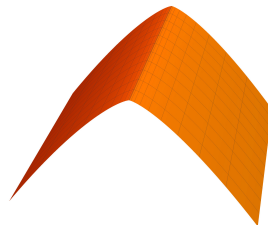
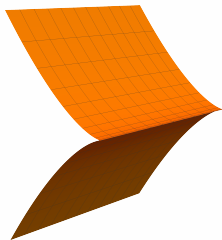
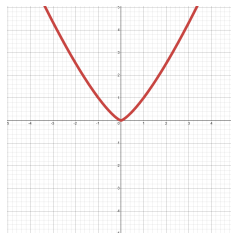
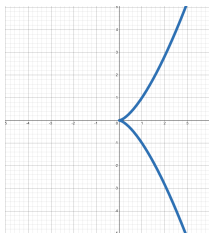
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イントロに、

property-A を整理したい。
というモチベーションを書く。

Cusp and (3,4)-cusp



- They look very similar at first glance.
- However, their geometric properties are quite different.

Introduction: Why $(3,4)$ -cuspidal edges?

- $(3,4)$ -cuspidal edges arise naturally on **parallel surfaces**.
- Despite their natural appearance, their geometry is less understood than that of ordinary cuspidal edges.

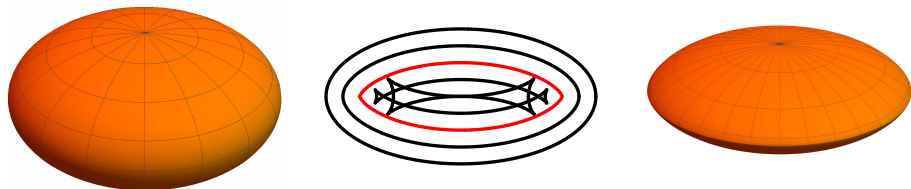


Figure: An ellipsoid (left), a parallel curve with a $(3,4)$ -cusp (center, **red**), and a parallel surface of the ellipsoid (right).

Introduction: Difference from cuspidal edges

Invariant and intrinsic invariant

In this talk,

- **Invariant**: its absolute value is preserved under coordinate changes and Euclidean isometries.
- **Intrinsic invariant**: determined only by the **first fundamental form**.

Cuspidal edge

(Saji–Umehara–Yamada (2009))

The singular curvature κ_s is an **intrinsic invariant**, and its sign is well-defined.

$\text{sgn } \kappa_s$ $\longleftrightarrow$ local shape



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$\text{sgn } \kappa_s$ $\longleftrightarrow$ local shape

(3, 4)-cuspidal edge

The sign of κ_s depends on the orientation of the null vector field:

$$\eta \longmapsto -\eta \implies \kappa_s \longmapsto -\kappa_s.$$

Hence its sign is **not well-defined**.



For $(3,4)$ -cuspidal edges, the sign of κ_s is **NOT** well-defined.

Question

Can we construct intrinsic invariants that describe the geometry of $(3,4)$ -cuspidal edges?

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Gaussian curvature

When is K bounded?

Shape

Convex or saddle?

Singularity type

Can we characterize $(3, 4)$ and $(3, 5)$?

Main results at a glance

We introduce new intrinsic invariants β, δ .

First fundamental form ds^2

$\implies$

Intrinsic data β, δ

Gaussian curvature

K bounded
 $\iff \beta \equiv \delta \equiv 0$

Local shape

$\operatorname{sgn} \beta(p)$
 $\implies$ convex / saddle

Singularity type

β, δ are zero or not
 $\implies (3, 4) / (3, 5)$

Generic $(3, 4)$ and $(3, 5)$ are preserved by isometries.

Basic setting

Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a smooth map, Σ is a two-dimensional manifold.

Frontal

f is a frontal $\iff \exists \nu : \Sigma \rightarrow \mathbb{S}^2$ s.t. $\langle f_u, \nu \rangle = \langle f_v, \nu \rangle = 0$.

Singular point

p is singular $\iff \text{rank}(df_p) < 2$.

Hence,

$$S(f) := \{ q \in \Sigma \mid \text{rank}(df_q) < 2 \}$$

denotes the **singular set**.

Let $\lambda := \det(f_u, f_v, \nu)$ be the signed area density function.

Basic setting

Let $p \in S(f)$ be a rank-1 singular point. Assume that $S(f)$ is a regular curve near p .

Singular and null vector fields

- A **singular vector field** ξ is tangent to the singular set:

$$\xi_p \in T_p S(f) \setminus \{0\}.$$

- A **null vector field** η satisfies

$$\eta_p \in \ker(df_p) \setminus \{0\}.$$

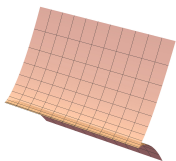
First and second-kind

First-kind : $T_p S(f) \neq \ker(df_p)$ e.g. $((3,4)-)$ cuspidal edge,

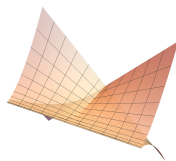
Second-kind : $T_p S(f) = \ker(df_p)$ e.g. swallowtail.

Examples of first- and second-kind singularities

First-kind singularities

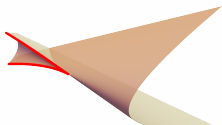


Cuspidal edge

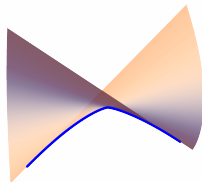


Cuspidal cross cap

Second-kind singularities at $p = (0, 0)$



Swallowtail



Cuspidal butterfly

3-type edges: a class containing (3, 4)-cuspidal edges

Let Σ be a two-dimensional manifold.

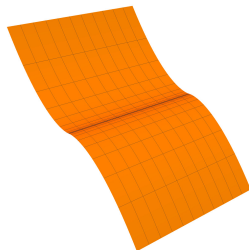
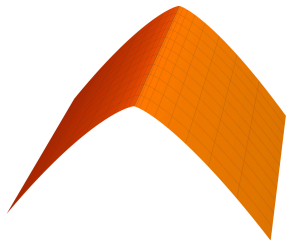
Definition(3-type edges) Martins–Saji–Santos–Teramoto (2024)

A smooth map $f : \Sigma \rightarrow \mathbb{R}^3$ has a **3-type edge** at $p \in \Sigma$ if the map-germ $f : (\Sigma, p) \rightarrow (\mathbb{R}^3, f(p))$ is $\mathcal{A}$ -equivalent to

$$(u, v) \longmapsto (u, v^3, v^4 \rho(u, v))$$

for some smooth function ρ .

(u, v^3, v^4) : (3, 4)-cuspidal edge, (u, v^3, v^5) : (3, 5)-cuspidal edge.



3-type edges: a class containing (3, 4)-cuspidal edges

Relation to generalized cuspidal edges

- For $m = 2$, m -type edge is exactly a **generalized cuspidal edge**:
(Honda–Naokawa–Saji–Umehara–Yamada (2020))

$$(u, v) \mapsto (u, v^2, v^3 \rho(u, v)).$$

- Thus, 3-type edge can be viewed as a **higher-order analogue**:

$$(u, v) \mapsto (u, v^3, v^4 \rho(u, v)).$$



Degeneracy order of the signed area density

Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a frontal with a unit normal vector field ν , and let $\lambda = \det(f_u, f_v, \nu)$ be the signed area density function.

New in this work

Definition (singular point of degeneracy order n)

Let $n \geq 1$. A rank-1 singular point p has **degeneracy order n** if there exist smooth real-valued functions $\hat{\lambda}$ and α such that

$$\lambda = \alpha \hat{\lambda}^n, \quad d\hat{\lambda}(p) \neq 0, \quad \alpha(p) \neq 0.$$

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- Degeneracy order $n \implies$ **singular set $S(f)$ is a regular curve** near p .
- $n = 1$: the usual non-degenerate case.
- We focus on $n = 2$.

Proposition

Let p be a rank 1 singular point of a frontal $f : \Sigma \rightarrow \mathbb{R}^3$. Then

p is of the first kind and
 p has degeneracy order 2 $\iff f$ has a 3-type edge at p .

Thus, 3-type edges can be viewed intrinsically as
first kind singularities of degeneracy order 2.

In this talk, we write $ds^2 = E du^2 + 2F du dv + G dv^2$, $G_{vj} := \frac{\partial^j G}{\partial v^j}$.

Normalized adapted coordinate system (NACS)

A local coordinate system (u, v) is called a **NACS** if

- $S(f) = \{v = 0\}$,
- $f_v(u, 0) = f_{v^2}(u, 0) = \mathbf{0}$,
- $\{f_u(u, 0), f_{v^3}(u, 0), \nu(u, 0)\}$ is an orthonormal basis of $\mathbb{R}^3$ and

$$\begin{aligned} E(u, 0) &= 1, & F(u, v) &\equiv 0, \\ G(u, 0) &= G_v(u, 0) = G_{vv}(u, 0) = G_{v^3}(u, 0) = 0, \\ G_{v^4}(u, 0) &= 6. \end{aligned}$$

Existence

A NACS exists near every singular point of the first kind of degeneracy order 2.

Intrinsic invariants β and δ

The higher derivatives of E and G depend on the choice of NACS.

New in this work

Definition(β, δ)

Along $S(f) = \{v = 0\}$, define

$$\beta(u) := -E_{v^4} + \frac{G_{v^5}}{20} E_{v^3},$$

$$\delta(u) := -E_{v^5} + \frac{G_{v^5}}{12} E_{v^4} - \left(\frac{7G_{v^5}^2}{1200} - \frac{G_{v^6}}{30} \right) E_{v^3}.$$

(All terms on the right are evaluated at $(u, 0)$.)

For another NACS,

$$\widehat{\beta} = \beta, \quad \widehat{\delta} = \pm\delta.$$

So, β and δ are **intrinsic invariants**.

What do β and δ measure?

Let K be the Gaussian curvature on the regular set.

Expansion of $v^2 K$

In a NACS,

$$v^2 K = \frac{1}{3}\beta + \left(\frac{1}{6}\delta - \frac{G_{v^5}}{360}\beta \right) v + \mathcal{O}(v^2).$$

$\beta \neq 0$

At a fixed point of the singular set,

$$K = \frac{\beta}{3v^2} + \mathcal{O}(v^{-1}).$$

β controls the **leading divergence**.

$\beta \equiv 0$

If $\delta \neq 0$, then

$$K = \frac{\delta}{6v} + \mathcal{O}(1).$$

δ controls the **next divergence**.

Main result I: boundedness of Gaussian curvature

Theorem A

Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a singular point of the first-kind with degeneracy order 2.

Then the following conditions are equivalent:

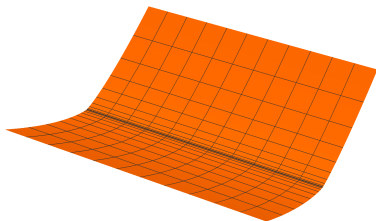
- (i) There exists a neighborhood U of p such that K is bounded on $U \setminus S(f)$.
- (ii) There exists a neighborhood U of p such that

$$\boxed{\beta \equiv 0, \quad \delta \equiv 0 \quad \text{on } S(f) \cap U.}$$

The first two coefficients of $v^2 K$ give a complete criterion for boundedness.

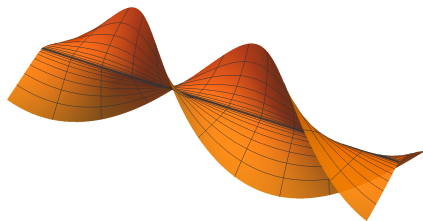
Main result I: examples

Both examples satisfy $\beta \equiv 0, \quad \delta \equiv 0.$



$$f_0(u, v) = \left(u, \frac{v^3}{6}, \frac{v^4}{24} \right)$$

$$K \equiv 0$$



$$f_{-1}(u, v) = \left(u - \frac{v^7}{1008}, \right. \\ \left. \frac{v^3}{6} \cos u - \frac{v^4}{24} \sin u, \right. \\ \left. \frac{v^3}{6} \sin u + \frac{v^4}{24} \cos u \right)$$

$$K \longrightarrow -1 \quad (v \rightarrow 0)$$

Main result II: intrinsic detection of the local shape

Generic at p

Let $\gamma(0) = p$ parametrize $S(f)$, and set $\hat{\gamma} := f \circ \gamma$.

p is **generic** if the osculating plane of $\hat{\gamma}$ exists and differs from LT_p .
Equivalently, for a first-kind singular point,

$$\kappa_\nu(p) := \frac{\langle \hat{\gamma}''(0), \nu(p) \rangle}{\|\hat{\gamma}'(0)\|^2} \neq 0.$$

Convex and saddle-shaped

Let LT_p be the limiting tangent plane at p .

f is **convex at p** $\iff \exists$ neighborhood V of p such that

$$f(V) \cap LT_p = \{f(p)\},$$

and $f(V) \setminus \{f(p)\}$ lies on one side of LT_p .

Otherwise, f is **saddle-shaped at p** .

Main result II: intrinsic detection of the local shape

Proposition B

Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a frontal, and let $p \in \Sigma$ be a generic $(3,4)$ -cuspidal edge. Then

- f is convex at $p \iff \beta(p) > 0$
- f is saddle-shaped at $p \iff \beta(p) < 0$

This is a pointwise statement at the fixed point p .

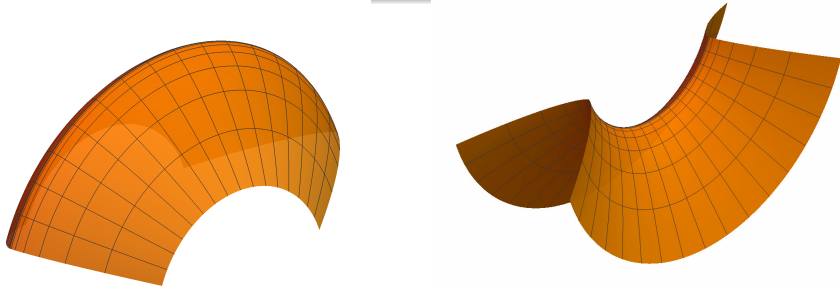


Figure: Local shapes at p : $\beta(p) > 0$ (left: convex) and $\beta(p) < 0$ (right: saddle-shaped).

Main result II: examples

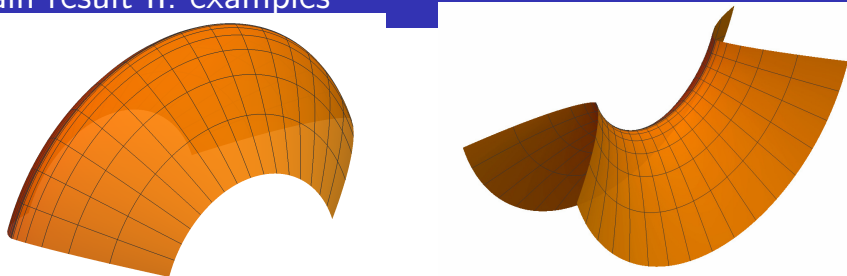


Figure: Local shapes at p : $\beta(p) > 0$ (left: convex) and $\beta(p) < 0$ (right: saddle-shaped).

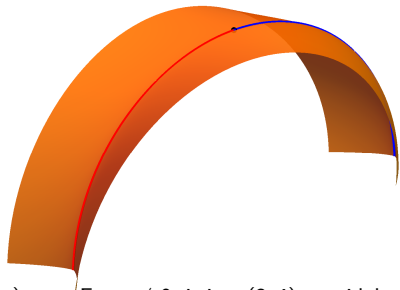


Figure: A surface with $\beta(u) = u$. For $u \neq 0$, it is a $(3, 4)$ -cuspidal edge; it is convex for $u < 0$ and saddle-shaped for $u > 0$.

From cuspidal curvatures to intrinsic data

For 3-type edges, Martins–Saji–Santos–Teramoto (2024) introduced the cuspidal curvatures

$$\omega_{3,4}, \quad \omega_{3,5}.$$

These quantities measure higher-order cuspidal behavior.

- $\omega_{3,4}(p)$ is an invariant.
- If $\omega_{3,4} \equiv 0$ along the singular set, then $\omega_{3,5}(p)$ is also an invariant.

Key relations

Under the genericity assumption,

$$\beta(p) \neq 0 \iff \omega_{3,4}(p) \neq 0.$$

If $\beta \equiv 0$ along the singular set near p , then

$$\delta(p) \neq 0 \iff \omega_{3,5}(p) \neq 0.$$

Criteria for $(3, 4)$ - and $(3, 5)$ -cuspidal edges

New in this work

Criterion for a $(3, 4)$ -cuspidal edge

Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a frontal and $p \in \Sigma$. Then

$$p \text{ is a } (3, 4)\text{-cuspidal edge} \iff \begin{cases} p \text{ is a 3-type edge,} \\ \omega_{3,4}(p) \neq 0 \end{cases}$$

Criterion for a $(3, 5)$ -cuspidal edge

Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a frontal and $p \in \Sigma$. Then

$$p \text{ is a } (3, 5)\text{-cuspidal edge} \iff \begin{cases} p \text{ is a 3-type edge} \\ \omega_{3,4} \equiv 0 \quad \text{on } S(f) \text{ near } p, \\ \omega_{3,5}(p) \neq 0. \end{cases}$$

These criteria are not intrinsic.

Main result III: intrinsic criteria for singularity types

Theorem C

Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a frontal, and let $p \in S(f)$ be a singular point of the first-kind with degeneracy order 2. Assume that f is generic at p . Then

$$f \text{ is a } (3,4)\text{-cuspidal edge at } p \iff \beta(p) \neq 0.$$

Moreover,

$$f \text{ is a } (3,5)\text{-cuspidal edge at } p \iff \begin{cases} \beta \equiv 0 & \text{on } S(f) \text{ near } p, \\ \delta(p) \neq 0. \end{cases}$$

- $\omega_{3,5}$ is defined under the assumption $\omega_{3,4} \equiv 0$ along $S(f)$.
- δ is defined without assuming $\omega_{3,4} \equiv 0$.

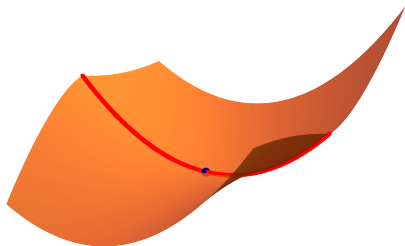
Main result III: examples

(3, 4)-cuspidal edge

$$f_{34}(u, v) = \left(u, v^3, \frac{u^2}{2} + uv^4 + v^5 \right).$$

For $p = (u_0, 0)$ with $u_0 \neq 0$,

$$\boxed{\beta(p) \neq 0}.$$

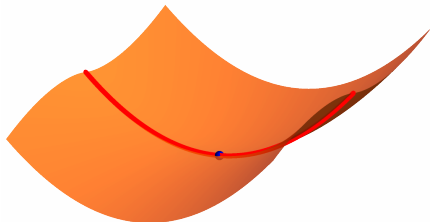


(3, 5)-cuspidal edge

$$f_{35}(u, v) = \left(u, v^3, \frac{u^2}{2} + v^5 \right).$$

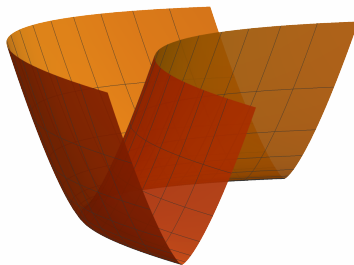
For $p = (u_0, 0)$,

$$\boxed{\beta \equiv 0 \text{ on } S(f), \quad \delta(p) \neq 0}.$$



Why genericity is needed

- The assumption $\kappa_\nu(p) \neq 0$ is essential. ($\kappa_\nu(p) = 0 \implies \beta(p) = 0$)
- Without it, $\beta(p)$ does not characterize $(3,4)$ -cuspidal edges.



$$f(u, v) = \left(u, \frac{u^2}{2} + \frac{v^3}{6}, \frac{v^4}{24} \right)$$

$(3,4)$ -cuspidal edge, but $\kappa_\nu \equiv 0$ and $\beta \equiv 0$.

Corollary: preservation of $(3, 4)$ - and $(3, 5)$ -cuspidal edges

For generic singular points of the first kind with degeneracy order 2,

$$\begin{aligned} \boxed{\text{same } ds^2} &\implies \boxed{\text{same intrinsic data } \beta, |\delta|} \\ &\implies \boxed{\text{being } (3, 4) \text{ or } (3, 5) \text{ is preserved by isometries}} \end{aligned}$$

Corollary D

Let

$$f : \Sigma \rightarrow \mathbb{R}^3, \quad g : \bar{\Sigma} \rightarrow \mathbb{R}^3$$

be locally isometric frontals near corresponding generic singular points p and $\bar{p}$ of the first kind and degeneracy order 2.

Then, for $j = 4, 5$,

$$f \text{ is a } (3, j)\text{-cuspidal edge at } p \iff g \text{ is a } (3, j)\text{-cuspidal edge at } \bar{p}.$$

New constructions

- Degeneracy order n for rank-1 singular points.
- NACS for first-kind singularities of degeneracy order 2.
- Intrinsic data β and δ from the first fundamental form.

Summary and outlook

New constructions

- Degeneracy order n for rank-1 singular points.
- NACS for first-kind singularities of degeneracy order 2.
- Intrinsic data β and δ from the first fundamental form.

Main results

- (i) K is bounded $\iff \beta \equiv \delta \equiv 0$.
- (ii) For a generic $(3,4)$ -cuspidal edge,

$$\beta(p) > 0 \iff \text{saddle-shaped}, \quad \beta(p) < 0 \iff \text{convex}.$$

- (iii) β, δ intrinsically characterize generic $(3,4)$ - and $(3,5)$ -cuspidal edges.
- (iv) Isometries preserve these two singularity types in the generic class.

Outlook

- Intrinsic classification of non-generic 3-type edges.
- Geometric meaning of δ .

Appendix: Singular and null directions

Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a frontal. Assume that $S(f)$ is a regular curve consisting of rank 1 singular points near $p \in S(f)$.

Let $\gamma(t)$ parametrize $S(f)$ with $\gamma(0) = p$.

Definition Kokubu–Rossman–Saji–Umehara–Yamada (2005)

- The one-dimensional subspace

$$T_{\gamma(t)}S(f) = \text{span}\{\gamma'(t)\}$$

is called the **singular direction**.

- The one-dimensional subspace

$$\ker(df)_{\gamma(t)}$$

is called the **null direction**. A nonzero vector $\eta(t)$ satisfying

$$df_{\gamma(t)}(\eta(t)) = 0$$

is called a **null vector**.

Appendix: m -type edges are frontals

Fact

Every m -type edge is a frontal.

An m -type edge is $\mathcal{A}$ -equivalent to

$$g(u, v) = (u, v^m, v^{m+1}\alpha(u, v)).$$

Set

$$\tilde{\nu} = \left(-mv^{m+1}\alpha_u, -v\{(m+1)\alpha + v\alpha_v\}, m \right).$$

Then,

$$\langle g_u, \tilde{\nu} \rangle = \langle g_v, \tilde{\nu} \rangle = 0, \quad \tilde{\nu} \neq 0.$$

Therefore,

$$\nu := \frac{\tilde{\nu}}{\|\tilde{\nu}\|}$$

is a smooth unit normal vector field along g .

m -type edge	$\implies$	frontal (preserved by $\mathcal{A}$ -equivalent)
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Appendix: Intrinsic nature of the assumptions

Let

$$ds^2 = E du^2 + 2F du dv + G dv^2.$$

- **First kind:** the null direction and the singular set $S(f)$ are determined by ds^2 . Hence

$$\eta(p) \notin T_p S(f)$$

is intrinsic.

- **Degeneracy order 2:** Since

$$\lambda^2 = EG - F^2,$$

the transverse vanishing order of λ along the regular singular curve is determined by ds^2 .

Both conditions are intrinsic.

Appendix

- Already for 4-type edges, the singularity type need not be preserved by isometries.
- Thus, it is nontrivial to ask whether this happens for **3-type edges**.



$(4,5)$ -cuspidal edge $(\sim_{\mathcal{A}}(u, v^4, v^5))$



$(4,5,+7)$ -cuspidal edge $(\sim_{\mathcal{A}}(u, v^4, v^5 + v^7))$

Figure: Isometric surfaces with **different singularity types**.
(Hattori–Honda–Morimoto (2025))

Appendix: Limiting normal curvature

Definition (limiting normal curvature and genericity)

Saji–Umehara–Yamada(2009); generalized by Martins–Saji–Umehara–Yamada(2016)

Let $f : \Sigma \rightarrow \mathbb{R}^3$ be a frontal, ν a unit normal vector field along f , and $p \in S(f)$ a singular point of the first kind.

Take an adapted coordinate system (u, v) such that

$$S(f) = \{v = 0\}, \quad f_u(u, 0) \neq 0.$$

The **limiting normal curvature** is defined by

$$\kappa_\nu(u) := \frac{\langle f_{uu}(u, 0), \nu(u, 0) \rangle}{\|f_u(u, 0)\|^2}.$$

In particular, if

$$\kappa_\nu(p) \neq 0,$$

we say that f is **generic at p** .

Appendix: Signed limiting normal curvature

Let ξ be a singular vector field and η a null vector field satisfying

$$\eta f = \eta^2 f = \mathbf{0} \text{ on } S(f), \quad \xi f, \eta^3 f \text{ linearly independent on } S(f).$$

New in this work

Definition

For a 3-type edge, define

$$\tilde{\kappa}_\nu(p) := \frac{\det(\xi f, \eta^3 f, \xi^2 f)}{\|\xi f \times \eta^3 f\| \|\xi f\|^2}(p).$$

- $\tilde{\kappa}_\nu$ is independent of the choice of a positively oriented pair (ξ, η) .
- It is the ordinary limiting normal curvature with the normal orientation fixed by (ξ, η) :

$$\tilde{\kappa}_\nu = \pm \kappa_\nu.$$

Appendix: Cuspidal curvatures

Let ξ be a singular vector field and η a null vector field such that

$$\eta f = \eta^2 f = \mathbf{0} \text{ on } S(f), \quad \xi f, \eta^3 f \text{ linearly independent on } S(f).$$

Definition ($(m, m + i)$ -cuspidal curvatures)

Martins–Saji–Santos–Teramoto (2024)

The $(3, 4)$ -cuspidal curvature is defined by

$$\omega_{3,4}(p) := \frac{\|\xi f(p)\|^{4/3} \det(\xi f(p), \eta^3 f(p), \eta^4 f(p))}{\|\xi f(p) \times \eta^3 f(p)\|^{7/3}}.$$

The $(3, 5)$ -cuspidal curvature is defined by

$$\omega_{3,5}(p) := \frac{\|\xi f(p)\|^{5/3} \det(\xi f(p), \eta^3 f(p), \eta^5 f(p))}{\|\xi f(p) \times \eta^3 f(p)\|^{8/3}}.$$

- $\omega_{3,4}(p)$ is an invariant (NOT intrinsic).
- If $\omega_{3,4} \equiv 0$ along the singular set, then $\omega_{3,5}(p)$ is also an invariant.

Appendix: Product curvatures and β, δ

The cuspidal curvatures $\omega_{3,4}$ and $\omega_{3,5}$ were introduced by Martins–Saji–Santos–Teramoto.

New in this work

Definition (product curvatures in this work)

$$\Pi_{3,4} := \tilde{\kappa}_\nu \omega_{3,4}, \quad \Pi_{3,5} := \tilde{\kappa}_\nu \omega_{3,5}.$$

$\Pi_{3,5}$ is well-defined when $\omega_{3,4} \equiv 0$ along the singular set.

Relation with β, δ

$$\Pi_{3,4} = -\frac{1}{2}\beta,$$

$$\beta \equiv 0 \text{ \& generic} \implies \Pi_{3,5} = -\frac{1}{2}\delta.$$

δ is defined without assuming $\omega_{3,4} \equiv 0$.

Appendix: Transformation formulas between NACSs

Let (u, v) and (ξ, η) be two NACSs near p , and write the coordinate transformation as $u = u(\xi, \eta)$, $v = v(\xi, \eta)$.

Let $\hat{E}$ and $\hat{G}$ denote the coefficients of the first fundamental form with respect to (ξ, η) .

Higher-order transformation formulas between NACSs

Along the singular set $\{v = 0\}$ (equivalently, $\{\eta = 0\}$),

$$\hat{E}_{\eta^3} = v_{\eta} E_{v^3},$$

$$\hat{E}_{\eta^4} = E_{v^4} + 6v_{\eta^2} E_{v^3},$$

$$\hat{E}_{\eta^5} = v_{\eta} E_{v^5} + 10v_{\eta^2} v_{\eta} E_{v^4} + (10v_{\eta^3} + 15\{v_{\eta^2}\}^2 v_{\eta}) E_{v^3},$$

$$\hat{G}_{\eta^5} = v_{\eta} (G_{v^5} + 120v_{\eta^2}),$$

$$\hat{G}_{\eta^6} = G_{v^6} + 27v_{\eta^2} G_{v^5} + 300v_{\eta} v_{\eta^3} + 1170\{v_{\eta^2}\}^2.$$

Here $\hat{E}$, $\hat{G}$, and the derivatives of v are evaluated at $(\xi, 0)$, while E and G are evaluated at $(u(\xi, 0), 0)$.

Appendix: A sign change of β in the generic case

The surface is generic along the whole singular set, but the origin is not a $(3,4)$ -cuspidal edge.

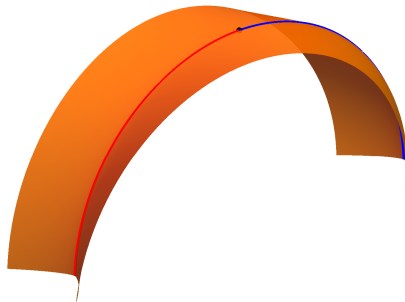


Figure: A surface with $\beta(u) = u$. It has a $(3,4)$ -cuspidal edge for $u \neq 0$.

Appendix: A sign change of β at a non-generic point

The surface has a $(3,4)$ -cuspidal edge at every point of the singular set, but the origin is non-generic.

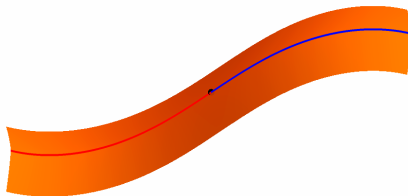


Figure: A $(3,4)$ -cuspidal edge with $\beta(u) = u$. The origin is non-generic.

Appendix: A non-generic example with a non-planar singular curve

Let the singular set be $S(f) = \{v = 0\}$.

An example that is non-generic along the singular set and whose singular curve is **non-planar**.

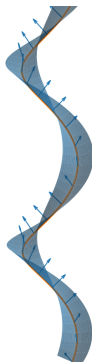


Figure: An example of a $(3,4)$ -cuspidal edge with $\beta = 0$.

Appendix: Map formulas for β and δ

Let (u, v) be a local coordinate system satisfying

$$S(f) = \{v = 0\}, \quad f_v(u, 0) = f_{vv}(u, 0) = \mathbf{0}, \quad \text{rank}(f_u, f_{v^3})(u, 0) = 2.$$

For δ , we fix the orientation of ∂_v .

Map formulas for β and δ

The quantities β and δ , originally defined in a NACS, can be written as follows:

$$\beta(u) = -2 \frac{\det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^4})}{\|f_u\|^{2/3} \|f_u \times f_{v^3}\|^{10/3}}(u, 0)$$

$$\begin{aligned} \delta(u) = & -2 \frac{\det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^5})}{\|f_u\|^{1/3} \|f_u \times f_{v^3}\|^{11/3}}(u, 0) \\ & + \frac{10}{3} \frac{\|f_u\|^{5/3}}{\|f_u \times f_{v^3}\|^{17/3}} \left\{ \langle f_{v^3}, f_{v^4} \rangle - \frac{\langle f_u, f_{v^3} \rangle \langle f_u, f_{v^4} \rangle}{\|f_u\|^2} \right\} \det(f_u, f_{v^3}, f_{uu}) \det(f_u, f_{v^3}, f_{v^4})(u, 0) \\ & + \frac{4}{3} \frac{\|f_u\|^{5/3}}{\|f_u \times f_{v^3}\|^{17/3}} \left\{ \langle f_{uu}, f_{v^3} \rangle - \frac{\langle f_u, f_{uu} \rangle \langle f_u, f_{v^3} \rangle}{\|f_u\|^2} \right\} \{\det(f_u, f_{v^3}, f_{v^4})\}^2(u, 0) \end{aligned}$$

Appendix

Counterexample to Theorem C without the genericity assumption:

$$f(u, v) = \left(u, \frac{u^2}{2} + \frac{v^3}{6}, \frac{v^4}{24} \right)$$

- The singular set is $\{v = 0\}$.
- Along the singular set, $\kappa_\nu = 0$ ($\implies \beta = \mathbf{0}$), but f is a $(3, 4)$ -cuspidal edge.

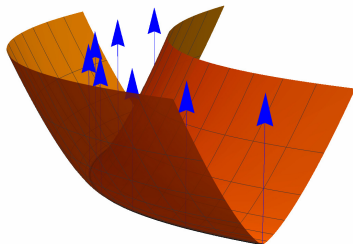


Figure: Counterexample to Theorem C in the non-generic case

Appendix

Singularity type is not necessarily preserved by isometries
(Hattori–Honda–Morimoto (2025)):



$(4, 5)$ -cuspidal edge



$(4, 5, +7)$ -cuspidal edge

Figure: Isometric surfaces with the same first fundamental form but different